

Raushan Singh

Senior Data Engineer

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Summary of Experience:

- Over 10 Years of strong experience as a Senior Data Engineer in Big Data, Data Warehousing and Business Intelligence including Requirements Analysis, Design Specification and Testing as per Cycle in both Waterfall and Agile methodologies.
- Extensive experience in designing and developing scalable Big Data solutions and data warehouse models on large-scale distributed data platforms, driving key business analytics and performance measurement.
- Proficient in using core components of the Hadoop ecosystem including HDFS, YARN, MapReduce, Hive, Impala, Pig, Sqoop, HBase, Spark, Spark SQL, Kafka, Spark Streaming, Oozie, and Hue to build and manage distributed data systems.
- Expertise in Azure Cloud Services including Azure SQL Data Warehouse, Azure Databricks, Azure Data Lake, Azure Blob Storage, Azure Data Factory, and Azure SQL Server, with experience in Azure development and Power BI for real-time data analytics.
- Strong working knowledge of AWS Cloud Platform services like EC2, S3, VPC, ELB, IAM, DynamoDB, CloudFront, CloudWatch, Route 53, Elastic Beanstalk, Auto Scaling, and Security Groups, focusing on scalable and secure cloud architecture.
- Proficient in working with NoSQL databases such as MongoDB, Cassandra, Couchbase, and Redis to efficiently store and retrieve unstructured data for real-time analysis.
- Extensive hands-on experience with ETL methodologies for data extraction, transformation, and loading in enterprise wide ETL Solutions and Data Warehouse tools to support robust reporting and data analytics.
- Expertise in all aspects of data engineering, including data collection, extraction, cleaning, aggregation, mining, validation, and reporting within both data warehousing and data lake environments.
- Skilled in troubleshooting, debugging, and optimizing ETL/ELT processes using techniques like push-down optimization and partitioning to manage large-scale datasets efficiently.
- Snowflake data warehouse specialist, with experience in building and optimizing ETL pipelines, ensuring efficient data storage, retrieval, and query performance.
- Expertise in Apache Spark (including Spark Streaming and Spark SQL) with Scala and Kafka, handling multiple data formats on HDFS and integrating real-time data streaming for analytics.
- Extensive experience with NoSQL databases including MongoDB, PostgreSQL, HBase, and Cassandra, and writing optimized HiveQL queries for large-scale data analysis and extraction.
- Proficient in importing and exporting data using Sqoop from S3 to Relational Database systems and loading into Hive tables with partitioning for optimized storage and query efficiency.
- Strong focus on data security, implementing encryption, access controls, and best practices in AWS and Azure environments to ensure the protection of sensitive data.
- Skilled in using workflow management tools like Oozie to orchestrate complex data workflows, manage dependencies, and ensure the seamless execution of ETL processes.
- Expertise in Data Modeling using techniques like Dimensional Data Modeling, Star Schema, and Snowflake Modeling for robust and scalable data architectures.
- Hands-on experience with Teradata utilities such as BTEQ, Fast Load, Multiload, Fast Export, and Unix Shell scripting to streamline data ingestion and processing workflows.
- Designed and implemented graph schemas for social networking platforms, supporting dynamic relationships and user interactions using efficient data structures.
- In-depth knowledge of Database Concepts, Algorithm Design, SDLC, OLAP/OLTP systems, Data Marts, and Data Lakes, ensuring efficient data flow across various systems.
- Experience with Kafka and Flume for data ingestion, managing both streaming and batch processes to efficiently load large datasets into Data Lakes and Warehouses.
- Proficient in all stages of the SDLC, including Agile, Scrum, and Waterfall methodologies, with hands-on experience in requirement analysis, development, testing, and deployment.
- Implemented data security best practices and compliance measures for NoSQL databases, ensuring data protection and privacy in highly regulated environments.

Technical Skills:

Big Data Technologies	Kafka, Cassandra, Apache Spark, Spark Streaming, HBase, Flume, Impala, HDFS, MapReduce, Hive, Pig, Sqoop, Flume, Oozie, Zookeeper
Hadoop Distribution	Cloudera CDH, Apache, AWS, Horton Works HDP
Programming Languages	SQL, PL/SQL, Python, R, PySpark, Pig, Hive QL, Scala, Shell Scripting, Regular Expressions
Spark components	RDD, Spark SQL (Data Frames and Dataset), and Spark Streaming
Cloud Infrastructure	AWS, Azure, GCP
Databases	Oracle, Teradata, My SQL, SQL Server, NoSQL Database (HBase, MongoDB)
Scripting & Query Languages	Shell scripting, SQL
Version Control	CVS, SVN and Clear Case, GIT
Build Tools	Maven, SBT
Containerization Tools	Kubernetes, Docker, Docker Swarm
Reporting Tools	Junit, Eclipse, Visual Studio, Net Beans, Azure Databricks, UNIX Eclipse, Visual Studio, Net Beans, Junit, CI/CD, Linux, Google Shell, Unix, Power BI, SAS, and Tableau

Project Experience:

Senior Azure Data Engineer

Change Healthcare, Nashville, TN

August 2023 – Present

Responsibilities:

- Leveraged Python, Shell Scripting, and Bash to interface with on-premises data lakes, analyzing application programs and managing ETL processes using T-SQL to ensure accurate and consistent data transformation across multiple systems.
- Designed and implemented complex data models in alignment with industry standards, recommending tailored models to meet project needs while ensuring optimal performance.
- Utilized PySpark and Data Frames for efficient data transformations, aggregations, and machine learning tasks, processing large datasets in distributed environments to enhance performance and scalability.
- Transformed and processed large datasets using Azure Databricks, storing transformed data in Parquet format to optimize storage efficiency and query performance.
- Orchestrated ETL processes by designing and managing Azure Data Factory pipelines, performing bulk data transfers from relational databases to Azure Data Lake Gen2.
- Integrated Azure Blob Storage with Databricks to process large datasets using Spark Scala scripts and UDFs, enabling efficient data transformations.
- Designed and implemented real-time streaming solutions using Apache Kafka, enabling continuous data ingestion and processing.
- Developed Azure Functions to integrate Kafka with event-driven data workflows, automating processing tasks triggered by specific events for real-time data insights and decision-making.
- Managed complex ETL processes in Azure Data Factory, utilizing visual tools to build pipelines for data ingestion, transformation, and storage from various sources.
- Created event driven ETL workflows in Azure Data Factory by integrating Azure Functions and Logic Apps, automating business processes within data pipelines for immediate data handling.
- Crafted and enhanced Spark ETL applications on Azure Databricks, optimizing data transformation and analysis tasks using Python and Scala.
- Integrated Snowflake with Azure Data Factory to orchestrate data pipelines and automate ETL workflows, enabling efficient data movement from Azure Data Lakes to Snowflake.
- Led the successful migration of critical data systems to PostgreSQL, ensuring improved performance and scalability.
- Developed and optimized advanced PostgreSQL queries and stored procedures to enhance data processing efficiency.

- Leveraged Azure Functions and Logic Apps for event-driven data processing and integration workflows within Azure Data Factory pipelines.
- Designed scalable and fault-tolerant Kafka architectures on Azure Cloud, utilizing Azure VMs and managed Kafka services such as Azure HDInsight.
- Developed and maintained Power BI reports and dashboards to visualize and analyze data trends, patterns, and insights, enabling stakeholders to make informed decisions.
- Utilized Power BI's advanced features such as DAX (Data Analysis Expressions) for creating complex calculations and measures, deriving actionable insights from data.
- Integrated Python scripts into Power BI for advanced data preprocessing, cleansing, and analysis, leveraging Python's extensive libraries for statistical analysis and machine learning.
- Integrated Informatica PowerCenter with Azure services such as Azure Data Lake Storage and Azure SQL Database to streamline data movement and processing workflows.
- Optimized Snowflake data warehouse on Azure Cloud platform for efficient data storage and processing.
- Designed and developed Snowflake schemas, tables, and views to support complex analytical queries and reporting requirements.
- Implemented complex ETL workflows using Informatica PowerCenter to extract, transform, and load data from diverse sources into data warehouses.
- Optimized Informatica mappings and sessions to enhance data processing performance and reduce load times, ensuring timely availability of data for analytics.
- Hands-on experience with Informatica Cloud Integration services, including Cloud Data Integration, Cloud Application Integration, and Cloud B2B Gateway.
- Integrated Azure Key Vault for secure key management, Azure Disk Encryption to encrypt virtual machine disks at rest, and Azure Functions for automating data encryption and decryption processes within Snowflake workflows, ensuring robust data security.
- Implemented Azure Active Directory (AAD) for secure authentication and authorization, safeguarding access to Azure data services and resources.

Environment: Azure Data Factory, Azure Databricks, Data Pipelines, Azure Data lakes, Azure Active Directory, Azure Key Vault, Informatica, Python, Pyspark, Scala, Bash, HDFS, Hadoop, Snowflake, Yarn, MapReduce, Hive, Sqoop, Oozie, Kafka, Spark SQL, Spark Streaming, Eclipse, Oracle, Teradata, PL/SQL, UNIX Shell Scripting, Power BI.

AWS Data Engineer

US Bank, Minneapolis, MN

May 2021 – July 2023

Responsibilities:

- Developed Python-based Spark applications to perform complex ETL tasks, including data cleansing, event enrichment, aggregation, and de-normalization, preparing data for machine learning and reporting.
- Created and orchestrated ETL workflows using AWS Glue, enabling seamless extraction, transformation, and loading (ETL) of data from sources like S3, DynamoDB, and Oracle databases.
- Integrated Apache Kafka with Spark to stream real-time data, supporting dynamic price surging using machine learning algorithms for enhanced data-driven decision-making.
- Designed and implemented comprehensive data pipelines utilizing AWS services such as Lambda, Glue, and Step Functions, ensuring efficient orchestration of data movement and processing.
- Utilized Apache Airflow for orchestrating, scheduling, and monitoring ETL jobs, enabling the timely and efficient processing of large datasets in a scalable environment.
- Developed automated data ingestion pipelines with Python and AWS SDKs, transferring data from sources such as S3, DynamoDB, and Oracle into Redshift with high accuracy and timeliness.
- Configured DynamoDB Streams and AWS Lambda functions to capture real-time data changes and load them into Redshift, facilitating near real-time analytics and insights.
- Managed ETL processes using AWS Glue, migrating data from external sources (e.g., S3, ORC/Parquet/Text files) into Redshift, improving data accessibility and analytical capabilities.
- Built and scheduled ETL pipelines in Apache Airflow, leveraging DAGs and various Airflow operators to ensure reliable and timely data processing within an AWS environment.
- Designed scalable ETL workflows with AWS Glue, migrating campaign data from external sources into AWS Redshift, enhancing data accessibility for deeper analytics.
- Leveraged Databricks to design, build, and deploy scalable data pipelines, enabling efficient data ingestion and processing for large datasets, improving overall system performance.
- Utilized SSIS control flow tasks and data flow components to orchestrate and manage complex ETL workflows, optimizing data loading performance and ensuring data accuracy.

- Integrated Oracle databases into data pipelines, writing optimized SQL queries and scripts for data extraction, transformation, and loading (ETL), ensuring smooth data flow between systems.
- Expertise in writing complex SQL queries for data retrieval, manipulation, and aggregation from relational databases, particularly Oracle, supporting business intelligence and reporting needs.
- Implemented Docker Registry for local upload/download of Docker images from Docker Hub, creating Docker files to automate the process of capturing and using images.
- Integrated Jenkins with tools such as Maven (build tool), Git (repository), SonarQube (code quality verification), and Nexus (artifact repository) to implement CI/CD pipelines, scheduling builds to support continuous development and testing.
- Troubleshooted performance issues within Tableau Server and resolved report-related problems, optimizing performance and user experience.
- Collaborated with stakeholders to understand business requirements, developing interactive Tableau dashboards for data visualization and reporting to support informed decision-making.
- Designed and developed Tableau reports and dashboards, providing actionable insights that support strategic business decisions through clear data visualization.
- Implemented AWS IAM roles and policies to ensure fine-grained access control, improving data security and compliance with industry standards.
- Utilized AWS Key Management Service (KMS) to encrypt sensitive data both at rest and in transit, ensuring data integrity and confidentiality.

Environment: Python, Pyspark, Scala, Kafka, HBase, Docker, Kubernetes, AWS, EC2, S3, Lambda, Cloud Watch, Auto Scaling, EMR, AWS Key, IAM, GIT, Airflow, Redshift, Jenkins, ETL, Spark, Hive, Athena, Sqoop, Pig, Oozie, Spark Streaming, Scala, Python, Data pipelines, Dynamo DB, Databricks, Tableau, GIT Micro Services.

Data Engineer

State of NY, Albany, NY

October 2019 – April 2021

Responsibilities:

- Designed solutions to process high volume data stream ingestion, processing and low latency data provisioning using Hadoop Ecosystems Hive, Pig, Scoop and Kafka, Python, Spark, Scala, NoSQL.
- Designed and implemented high-volume data stream ingestion and processing solutions with low-latency provisioning using Hadoop Ecosystem tools such as Hive, Pig, Sqoop, and Kafka, along with programming in Python, Spark, and Scala, and working with NoSQL databases.
- Utilized Hive SQL, Presto SQL, and Spark SQL to query large datasets and develop ETL processes for loading and extracting data across distributed systems, optimizing data pipelines for enhanced performance and scalability.
- Developed interactive data visualizations using tools like Tableau and SAS Visual Analytics, publishing dashboards, reports, and workbooks to provide actionable insights.
- Designed and built a multi-terabyte Data Warehouse infrastructure from scratch on Amazon Redshift, handling millions of records daily for large-scale data processing and storage.
- Implemented big data ingestion pipelines using Kafka and Spark Streaming, performing data quality checks, transformations, and storing data in optimized formats, handling multi-terabyte datasets from various data sources for downstream analytics using Spark.
- Imported and transformed data from multiple sources using Hive and MapReduce, loading transformed data into HDFS for further processing and analysis.
- Performed system analysis for enhancements and new functionalities, including impact analysis of ETL changes to ensure smooth integration and minimal disruption to existing systems.
- Managed AWS security groups with a focus on high availability, fault tolerance, and auto-scaling using Terraform templates, along with Continuous Integration and Continuous Deployment (CI/CD) utilizing AWS Lambda and AWS CodePipeline.
- Developed SSRS reports and created SSIS packages for extracting, transforming, and loading data (ETL) from multiple source systems, ensuring timely and accurate data integration.
- Tuned SQL queries using tools like Execution Plan, Query Analyzer, SQL Profiler, and Database Engine Tuning Advisor to improve performance and optimize application workflows.
- Built complex SSIS/ETL packages for data extraction, transformation, and loading, ensuring data integrity and efficiency in processing across various systems.
- Designed and developed Snowflake schemas, tables, and views, supporting complex analytical queries and reporting requirements for data-driven decision-making.
- Participated in the full software development lifecycle (SDLC), collaborating on requirements gathering, solution design, development, testing, and product support within Agile (Scrum) and other iterative methodologies.

Environment: Oracle, Kafka, Python, Redshift, Informatica, ETL, Scala, AWS, EC2, S3, SQL Server, Erwin, RDS, NOSQL, MySQL, Dynamo DB, PostgreSQL, Tableau, Git Hub, SSIS, SSRS, SQL, PLSQL, Python, Pig.

Data Engineer

Costco, Issaquah, WA

May 2017 – September 2019

Responsibilities:

- Extracted, transformed, and loaded (ETL) data from various sources into the target SQL Server Database. Implemented Copy Activity and Custom Azure Data Factory Pipeline Activities for cloud based ETL processing.
- Migrated SQL databases to Azure Data Lake, Azure SQL Database, Databricks, and Azure SQL Data Warehouse, managing database access controls and migrating on-premises databases to Azure Data Lake Store using Azure Data Factory.
- Developed data pipelines using Sqoop to ingest customer behavioral data and purchase histories into HDFS for in-depth analysis.
- Transformed data in Azure Databricks, storing it in Parquet formats for optimized and efficient storage.
- Enhanced and optimized Spark code for aggregating, grouping, and executing data mining tasks using the Apache Spark framework.
- Imported real-time data into Hadoop using Apache Kafka and implemented Oozie jobs for daily data imports.
- Managed the complete big data flow of the application, starting from data ingestion into HDFS, followed by processing and analyzing data within the Hadoop ecosystem.
- Migrated SQL databases to Azure Data Lake, Azure SQL Database, and Databricks using Azure Data Factory, optimizing cloud data infrastructure for efficiency.
- Developed real-time streaming analytics pipelines, building connections between Event Hub and Azure Streaming Analytics for event-driven processing.
- Proficient in Azure Data Factory, creating dependencies between activities and orchestrating complex workflows for seamless data pipeline operations.
- Created partitioned and bucketed Hive tables in Parquet file formats with Snappy compression, loading data from Avro Hive tables into Parquet Hive tables for performance optimization.
- Imported data into HDFS using Sqoop from multiple RDBMS servers, and exported data back to RDBMS servers via Sqoop, ensuring smooth bi-directional data flow.
- Integrated Azure Key Vault for secure key management and Azure Disk Encryption for encrypting virtual machine disks at rest. Automated data encryption and decryption processes using Azure Functions within Snowflake workflows, ensuring robust data security in the Azure environment.

Environment: Azure, HDFS, Hadoop, Yarn, MapReduce, Hive, Sqoop, Oozie, Kafka, Spark SQL, Spark Streaming, Eclipse, Oracle, Teradata, PL/SQL UNIX Shell Scripting.

ETL Developer

Client: Value Labs, Hyderabad, India

June 2015 – Apr 2017

Responsibilities:

- Developed Python scripts using Pandas to standardize and manipulate data frames, ensuring consistent data quality and integrity throughout the data processing pipeline.
- Created and organized HDFS staging areas to streamline data ingestion and improve the efficiency of preliminary data processing.
- Designed HDFS structures for optimal data storage and quick retrieval, enhancing overall system performance and reducing data processing times.
- Managed the ETL process for migrating legacy data from SQL Server and Teradata into modern target systems, ensuring seamless integration and data quality during migration.
- Developed SQL scripts in Teradata and SQL Server Management Studio (SSMS) for data upload, retrieval, manipulation, and secure handling of sensitive data.
- Implemented UNIX scripts to define workflows and automate data processing tasks, ensuring efficiency and accuracy in data transformations.
- Developed a raw layer of external tables containing copied data from HDFS, organizing data for downstream processes.
- Created internal tables in Hive for data manipulation and organization within the data service layer, facilitating efficient data management and processing.
- Exported data into SQL Server by creating Staging Tables for loading and processing data, ensuring smooth data transfer and system compatibility.
- Conducted data comparison across multiple databases to ensure data integrity, identifying and resolving data quality issues during transformation and loading processes.
- Analyzed and investigated data quality to identify potential data loss or corruption during data loads, implementing preventive measures to ensure data accuracy.

Environment: HDFS, Python, Hadoop, Hive, HBase, MapReduce, Spark, SQL Server, PostgreSQL, Unix.

Educational Background:

Bachelors in CSE, 2015